

# Mask, moment time term, cross-bank near-copies

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Tags: [code] = read in the code; [measured] = number from a run or script; [hypothesis] = not verified.

## Notation

|                                                                                                                                        |                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x \in \mathbb{R}^M$                                                                                                                   | one signal ( $M = 256$ ); $x_k^i$ , $i = 1..N$ : the $N$ walkers at step $k$ , time $t_k$ ;<br>$\Delta t_k = t_{k+1} - t_k$                                         |
| $\phi = (\phi_1, \dots, \phi_r)$                                                                                                       | the $r = 281$ statistics, all families concatenated                                                                                                                 |
| $u \cdot v$                                                                                                                            | dot product in $\mathbb{R}^M$                                                                                                                                       |
| $G$                                                                                                                                    | Gram matrix on a walker set: $G_{nm} = \frac{1}{N} \sum_i \nabla \phi_n(x^i) \cdot \nabla \phi_m(x^i)$                                                              |
| $Gc = b$                                                                                                                               | generic form of the two systems solved at each step: predictor ( $c = \eta$ , $G$ at $x_k$ )<br>and corrector ( $c = \theta$ , $G$ at the predicted walkers $y_k$ ) |
| $D, \tilde{G}$                                                                                                                         | $D = \text{diag}(G)$ , $\tilde{G} = D^{-1/2} G D^{-1/2}$ (so $\tilde{G}_{nn} = 1$ )                                                                                 |
| $\varepsilon$                                                                                                                          | ridge ( <b>-regularization</b> ; $10^{-4}$ in the zfloor runs)                                                                                                      |
| The step solves $(\tilde{G} + \varepsilon I)\tilde{c} = D^{-1/2}b$ , $c = D^{-1/2}\tilde{c}$ , i.e. $(G + \varepsilon D)c = b$ [code]. |                                                                                                                                                                     |

## 1 Dead statistics and the mask

(applied)

**Definition** [code]. At step  $k$ , statistic  $n$  is *dead* if  $G_{nn} \leq \tau_n$ ; its coefficient is set to 0 and the system is solved for the others. With the live set  $L = \{n : G_{nn} > \tau_n\}$ :

$$c_n = 0 \ (n \notin L), \quad (G_{LL} + \varepsilon D_{LL}) c_L = b_L.$$

**Why.** If statistic  $n$  is nearly decoupled ( $G_{nm} \approx 0$ ,  $m \neq n$ ), then  $c_n \approx b_n / ((1 + \varepsilon)G_{nn})$ ; its term in the walker update is  $c_n \nabla \phi_n$ , of mean square  $c_n^2 G_{nn} \approx b_n^2 / G_{nn}$ . This diverges as  $G_{nn} \rightarrow 0$  unless  $b_n$  vanishes at least as fast as  $\sqrt{G_{nn}}$ . The ridge does not prevent it, since it is proportional to  $G_{nn}$ . Job 268295 (walker blow-up at step 163) was diagnosed as this on 28 Sep, and the rerun with the mask completed [measured].

**Thresholds** [code].  $\tau_n = 10^{-6}$  for the region statistics (**Scalar\_psi\_gaussianK**, **Scalar\_morlet\_gaussianK**); these are rescaled so that  $\mathbb{E}_{\text{data}} |\nabla \phi_n|^2 = 1$ , so this is  $10^{-6}$  of the data value.  $\tau_n = 0$  for all other statistics (dead only if exactly zero); these are not rescaled.

**Why only the region statistics.** A region statistic is [code]  $\phi_{j\ell}(x) = \frac{1}{s_{j\ell} M} \sum_u w_\ell(|z_j(u)|) |z_j(u)|^{\alpha_{j\ell}}$ , with  $z_j = \text{Re}(x \star \psi_j)$ ,  $\psi_j$  filter  $j$ ,  $|\cdot|$  slightly smoothed,  $s_{j\ell}$  the rescaling constant, and  $w_\ell$  a window selecting the amplitude range  $\ell$  with sigmoid edges of width  $s$ . For the outermost range,  $w_\ell(a) = (1 + e^{-(a-c)/s})^{-1} \approx e^{-(c-a)/s}$  when  $c - a \gg s$ : if all amplitudes are well below the cut  $c$ , the gradient is exponentially small but not zero. On the walkers  $G_{nn}$  reached  $10^{-16}$ – $10^{-8}$  for such regions (coarse channel, early  $t$ ) [measured, 28 Sep]. The threshold was added only where this was observed. For the other families nothing guarantees  $G_{nn}$  stays away from 0 [hypothesis that it does]. A possible exception:  $L_6^\psi$  [21] uses the pure mean filter

(frequency bin 0), so its gradient is  $\propto (\text{signal mean})^5$ , which is small on noise-like walkers (not checked). This is not monitored: run 382722 logs  $\min_n G_{nn}/\tau_n$  only over statistics with  $\tau_n > 0$ .

**In  $\Theta_{\text{reg}}$  [code].** Same mask:  $\Theta_n(t_k) = 0$  for  $n \notin L_k$  (row and column replaced by the identity, right-hand side 0).

## 2 Moment time term of $\Theta_{\text{reg}}$

(applied, default)

|                              |                                                                                                                                                                                                                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\theta_k$                   | corrector coefficients at step $k$ , divided by $\Delta t_k \sigma^2$ ( $\sigma$ : noise level)                                                                                                                                                                                                         |
| $\Theta(t) \in \mathbb{R}^r$ | smoothed coefficient path ( $\Theta_{\text{reg}}$ ), the unknown                                                                                                                                                                                                                                        |
| $p_t$                        | law of the walkers at time $t$ (no exponential form assumed)                                                                                                                                                                                                                                            |
| $m_t$                        | target moments, $m_t = \mathbb{E}[\phi(I_t)]$ , $I_t$ the interpolant; the corrector enforces $\mathbb{E}_{p_t}[\phi] \approx m_t$                                                                                                                                                                      |
| $\dot{m}_t$                  | $\frac{d}{dt} \mathbb{E}[\phi(I_t)] = \mathbb{E}[\nabla \phi(I_t) \cdot \dot{I}_t]$ (pathwise, already used by the predictor)                                                                                                                                                                           |
| $\Sigma_t$                   | $\text{Cov}_{p_t}(\phi)$ , centred, computed on the corrected walkers $x_{k+1}$                                                                                                                                                                                                                         |
| $M_k, b_k$                   | $M_k = G(y_k) + \varepsilon' \text{diag } G(y_k)$ , $b_k = (m_{t_{k+1}} - \frac{1}{N} \sum_i \phi(y_k^i)) / (\Delta t_k \sigma^2)$<br>with $\varepsilon' = \varepsilon$ , $M_k \theta_k = b_k$ is the corrector's equation [code]<br>( $\varepsilon'$ : RIDGE of the offline solve, default $10^{-4}$ ) |

**Time term.** Fit the time derivative of  $\log p_t$  with  $\dot{\Theta} \cdot \phi(x) + a$  ( $a \in \mathbb{R}$  free) in least squares:  $\min_{\dot{\Theta}, a} \mathbb{E}_{p_t}[(\partial_t \log p_t - \dot{\Theta} \cdot \phi - a)^2]$ . Eliminating  $a$  (using  $\mathbb{E}_{p_t}[\partial_t \log p_t] = 0$ ) gives

$$\Sigma_t \dot{\Theta} = \text{Cov}_{p_t}(\phi, \partial_t \log p_t) = \int \phi \partial_t p_t dx = \frac{d}{dt} \mathbb{E}_{p_t}[\phi] = \dot{m}_t.$$

The last step uses only  $\mathbb{E}_{p_t}[\phi] = m_t$ . The old (legacy) term was built from the noise  $Z$  of the interpolant, but the walkers are drawn independently of  $Z$  [code], so its conditional mean given the walkers is a constant [derived].

**Energy** ( $\lambda \geq 0$  chosen after the run):

$$\mathcal{E}[\Theta] = \int_0^1 \left[ \frac{1}{2} \Theta^\top M_t \Theta - b_t^\top \Theta \right] dt + \lambda \int_0^1 \left[ \frac{1}{2} \dot{\Theta}^\top \Sigma_t \dot{\Theta} - \dot{m}_t^\top \dot{\Theta} \right] dt.$$

$\lambda = 0$ :  $\Theta(t_k) = \theta_k$ . Large  $\lambda$ :  $\Sigma_t \dot{\Theta} \rightarrow \dot{m}_t$ , the data term fixing the rest. Stationarity:  $M_t \Theta - b_t = \lambda \frac{d}{dt} (\Sigma_t \dot{\Theta} - \dot{m}_t)$ .

**Discretisation** [code] (trapezoid weights  $w_k$ ): first integral  $\rightarrow \sum_k w_k [\cdot]_k$ ; second  $\rightarrow \sum_k \lambda \Delta t_k [\frac{1}{2} \delta_k^\top C_k \delta_k - \tilde{m}_k^\top \delta_k]$  with  $\delta_k = (\Theta_{k+1} - \Theta_k) / \Delta t_k$ ,  $C_k = \frac{1}{2} (\Sigma_k + \Sigma_{k+1})$ ,  $\tilde{m}_k = \frac{1}{2} (\dot{m}_k + \dot{m}_{k+1})$ . Solved exactly (block tridiagonal, float64).

## 3 Cross-bank near-copies

(found; fix proposed, not applied)

**Meaning.** With  $g_n = \nabla \phi_n / \sqrt{G_{nn}}$  (unit mean-square norm),  $\tilde{G}_{nm} = \frac{1}{N} \sum_i g_n(x^i) \cdot g_m(x^i)$  is the cosine similarity of the two gradients (uncentred), over walkers and positions. An eigenvector  $v \approx (e_a \mp e_b) / \sqrt{2}$  of  $\tilde{G}$  with eigenvalue  $\mu$  gives

$$\tilde{G}_{ab} \approx \pm(1 - \mu), \quad \frac{1}{N} \sum_i |g_a(x^i) \mp g_b(x^i)|^2 \approx 2\mu,$$

i.e. the two normalised gradients are equal up to a relative difference  $\sim \sqrt{2\mu}$ . The walker update contains  $\tilde{c}_a g_a + \tilde{c}_b g_b \approx (\tilde{c}_a \pm \tilde{c}_b) g_a$  ( $\tilde{c}_n = \sqrt{G_{nn}} c_n$ ), so the data fix only  $\tilde{c}_a \pm \tilde{c}_b$ . The component along  $v$  is  $v^\top \tilde{c} = v^\top \tilde{b} / (\mu + \varepsilon) \approx v^\top \tilde{b} / \varepsilon$ : its size is set by  $\varepsilon$ , not by  $\mu$ .

**Found** [measured; zfloor run, seed 900, `find_duplicate_statistics.py` on  $M_k$ , 25 nodes] (global index in brackets;  $S^{\text{morlet}}$ ,  $S^\psi = \text{Scalar\_morlet/psi\_gaussianK}$ ):

| pair                                                     | first seen   | $\mu$ when first seen as a separate pair |
|----------------------------------------------------------|--------------|------------------------------------------|
| $L_6[7]$ (7) $\approx L_6^\psi[20]$ (29)                 | $t = 0.0002$ | $6.9 \cdot 10^{-7}$                      |
| $L_6[8]$ (8) $\approx L_6^\psi[21]$ (30)                 | $t = 0.23$   | $6.2 \cdot 10^{-7}$ (at $t = 0.45$ )     |
| $S^{\text{morlet}}[30]$ (140) $\approx S^\psi[77]$ (109) | $t = 0.23$   | $5.8 \cdot 10^{-6}$ (at $t = 0.45$ )     |

At  $t = 0.23$  the last two pairs appear mixed in two 4-statistic directions ( $\mu = 1.0 \cdot 10^{-7}, 2.1 \cdot 10^{-6}$ ). The tool prints only the first appearance, so presence at all later times is not shown by it. Separately, the per-step  $\tilde{G}$  of the corrector has  $\mu_{\min} \approx 1\text{--}6 \cdot 10^{-7}$  at all steps with largest loadings on 7, 8, 29, giving  $\kappa(\tilde{G}) \sim 10^7\text{--}10^8$  [measured, offline job 382751].  $\kappa$  with these pairs removed has not been computed.

**Mechanism** [measured, `check_cross_bank.py`, synthetic signals].  $L_6$  and `Scalar_morlet` use the  $Q = 1$  bank (9 filters),  $L_6^\psi$  and `Scalar_psi` the  $Q = 3$  bank (25 filters, 22 kept by dedup; kept index 21 is bank channel 24). Filter deficit  $1 - |\langle f, g \rangle| / (\|f\| \|g\|)$  (Fourier domain):

| pair                        | content             | filter deficit      | $1 -  \tilde{G}_{ab} $ of $L_6$ grads (white / $k^{-2}$ ) |
|-----------------------------|---------------------|---------------------|-----------------------------------------------------------|
| $Q=1$ #7 vs kept $\psi$ #20 | bin 1               | $5.8 \cdot 10^{-4}$ | $2.4 \cdot 10^{-6} / 4.0 \cdot 10^{-7}$                   |
| $Q=1$ #8 vs kept $\psi$ #21 | bin 0 (#8: + bin 1) | $1.5 \cdot 10^{-3}$ | $1.2 \cdot 10^{-5} / 1.4 \cdot 10^{-5}$                   |

So the filters differ by  $\sim 10^{-3}$  but the gradients by only  $10^{-7}\text{--}10^{-5}$ , the same order as the  $\mu$  measured on the walkers. Likely reason: the gradient applies the filter a second time [hypothesis]. On these signals the dependence is a near-copy, not an exact one; on the walkers  $\mu$  sits close to float32 resolution, so walker values alone could not tell. Not checked: the filters behind the `Scalar` pair (requires the fitted state).

**Why dedup missed them** [code]. `deduplicate_filters` drops a filter whose deficit to an earlier one is  $\leq \text{tol}$  ( $10^{-8}$  in these runs), and it only compares filters *inside* the  $Q = 3$  bank. The two banks are never compared, and the deficits here ( $\sim 10^{-3}$ ) are far above  $10^{-8}$  anyway.

**Proposed fix.** Apply the same filter test between the banks and drop the  $Q = 1$  filter (for  $L_6$ , and for `Scalar_morlet` if it uses the same filter) when its deficit to a kept  $\psi$  filter is below  $10^{-2}$ . The gap is clean [measured]: the two copies are at  $5.8 \cdot 10^{-4}$  and  $1.5 \cdot 10^{-3}$ ; every other  $Q=1$  filter is  $\geq 7.4 \cdot 10^{-2}$  from its closest  $\psi$  filter; kept  $\psi$  filters are  $\geq 4.2 \cdot 10^{-2}$  from each other. This removes  $Q=1$  filters #7 and #8 only.